

Data Structures

Lecture 11

Hash Function



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Topics

- Hash Function
 - Introduction
 - Strong Hash Algorithm (Properties)
 - How Hash Works (Division Method)

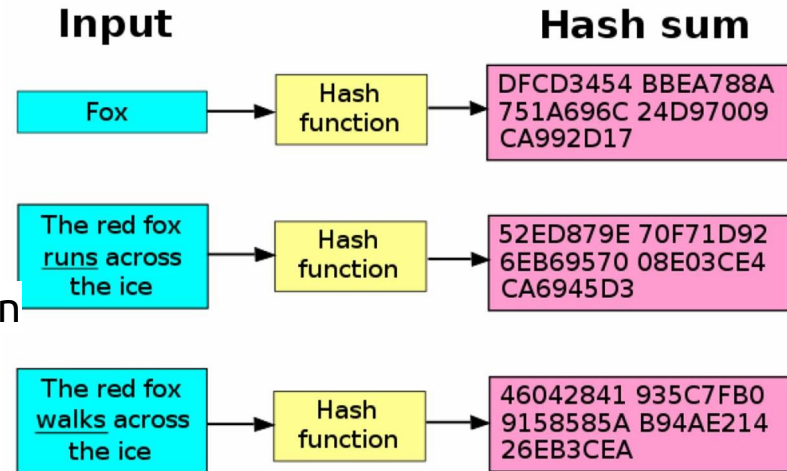
Hashing



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Hash - Introduction

- Hash function is an algorithm that takes a string "**message**" with **any size** and converts it into a string "**hash**" with a **fixed size**.
- Hash functions are **one-way** (we cannot get the input from the hashed output)
- Used for
 - Securely store passwords in a database,
 - Ensure data integrity (the data has not been changed)
- The same data produces the same hash.
We can validate the data (password) if we know its hash:
- The receiver uses the same hash function to generate the hash of the password and compares it to the stored hashed password. If they are the same, it is correct.



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Hash - Properties of a Strong Hash Algorithm

Different Hash Functions

Input (Message)	Hash Function	Output (Hash Value)
CFI	MD5 (128-bit, 16-byte) 32 characters	3A10 0B15 B943 0B17 11F2 E38F 0593 9A9A
CFI	SHA-1 (160-bit, 20-byte) 40 characters	569D C9F0 7B48 7F58 9241 AD4C 5C28 7DA0 A448 8D08
CFI	SHA-256 (256-bit, 32-byte) 64 characters	F3ED 0867 48FF 3641 3091 0BB6 6293 7080 2958 B5A2 52AF F364 1FC5 07FD E80D 9929

Hash functions must have these three properties for better security:

- **Collision-free:** no two different input values should map to the same output hash.
- **Hidden:** It should be difficult to guess the input value from the function's output.
- **Puzzle-friendly:** It must be difficult to select an input that provides a known output.

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Hash - How Hash Works

There are many hash functions that use numeric or alphanumeric keys:

1. Division Method:

The most simple method that divides the value k by M and then uses the remainder obtained.

Formula: $h(K) = k \bmod M$ k : value , M : size of the hash table

- $k = 12345, M = 95$
 $h(12345) = 12345 \bmod 95 = 90$
- $k = 1276, M = 11$
 $h(1276) = 1276 \bmod 11 = 0$

Limitation: consecutive keys map to consecutive hash values in the hash table.



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Hash - How Hash Works

2. Digit Folding Method

1. Divide the key-value **k** into a number of parts i.e. ***k1, k2, k3,...,kn***, where each part has the same number of digits except for the last part that can have lesser digits.
2. Add the individual parts. The hash value is obtained by ignoring the last carry if any.

Formula: $k = k_1, k_2, k_3, k_4, \dots, k_n \rightarrow h(K) = k_1 + k_2 + k_3 + k_4 + \dots + k_n$

- $k = 12345$

$$k_1 = 12, k_2 = 34, k_3 = 5$$

$$h(K) = k_1 + k_2 + k_3 = 12 + 34 + 5 = 51$$

